

Student 27

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Q1. String Character Frequency Counter

CODE

```
s = input()
s = s.lower()
count={}
for ch in s:
    if ch not in count:
        count[ch] = 1
    else:
        count[ch] += 1
for ch in sorted(count):
    print(ch,"->",count[ch])
```

INPUT hello world

OUTPUT (no output)

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s=input().lower()
v="aeiou"
vd={}
cd={}
vc=cc=0
for i in s:
    if i.isalpha():
        if i in v:
            vc+=1
            vd[i]=vd.get(i,0)+1
        else:
            cc+=1
            cd[i]=cd.get(i,0)+1
            print("vowels:",vc)
            print("consonants:",cc)
            if vd:
                x=max(vd,key=vd.get)
                print("Most frequent vowel: ",x,"(",vd[x],")")
            if cd:
                y=min(cd,key=cd.get)
                print("Least frequent consonant: ",y,"(",cd[y],")")
            print("vowel-to-consonant ratio:",round(vc/cc,2)if cc else 0)
```

INPUT python programming

OUTPUT (no output)

Q3. String Compression and Run-Length Encoding

CODE

```
s=input()
r=""
i=0
while i<len(s):
    c=s[i]
    while i<len(s)-1 and s[i]==s[i+1]:
        c+=1
        i+=1
    r+=s[i]+str(c)
    i+=1
    print("compressed:",r)
    d=""
    i=0
    while i<len(r):
        d+=i+int(r[i+1])
        i+=2
    print("decompressed:",d)
```

INPUT aabbccdddee

OUTPUT (no output)

Q4. Advanced Text Statistics Analyser

CODE

```
import re
from collections import Counter
text=input()
stop=["the","is","a","an","in","of","and","to","for"]
sent=len(re.findall('[!?',text])
if sent==0:
    sent=1
    para=1
    words=re.findall
```

OUTPUT (no output)